



**Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus**

Lab Rubrics Form	
Lab Name	Implementation of Array Control, MathScript Functions, and Temperature Conversion using LabVIEW
Lab Number	01
Submitted by:	
Student Names	Registration #
Sumaira Safeer	FA22-BCE-019

Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

LAB: 01

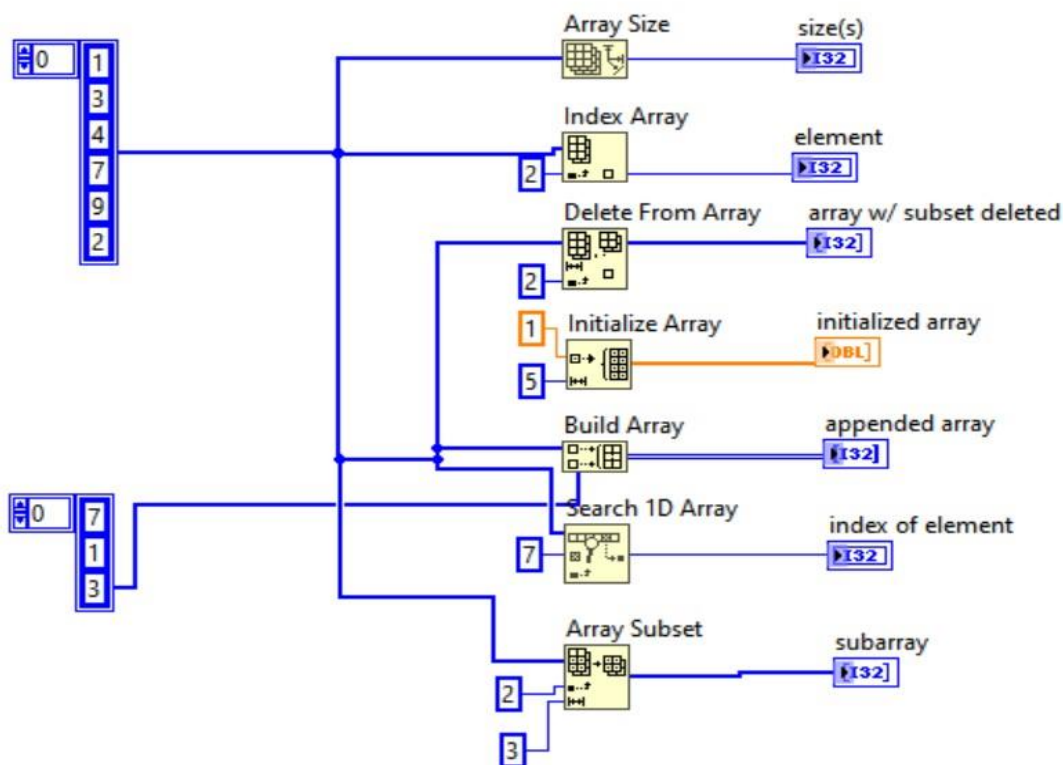
Implementation of Array Control, MathScript Functions, and Temperature Conversion

Objective

- To understand the use of Array Control for accepting user input in LabVIEW.
- To generate and plot a 10 Hz sine wave using both MathScript and MathScript Node.
- To learn the implementation of Case Structure and Enum for temperature conversion between Celsius and Fahrenheit.
- To develop basic programming skills in LabVIEW using graphical programming concepts and built-in functions.

Task 01:

Implement the array control as input instead of array constant.



size(s)
6

element
4

array w/ subset deleted
0 1 3 7 9

initialized array
0 1 1 1 1 1 0

appended array
0 1 3 4 7 9 2 0
0 7 1 3 0 0 0 0

subarray
0 4 7 9 0 0

index of element
3

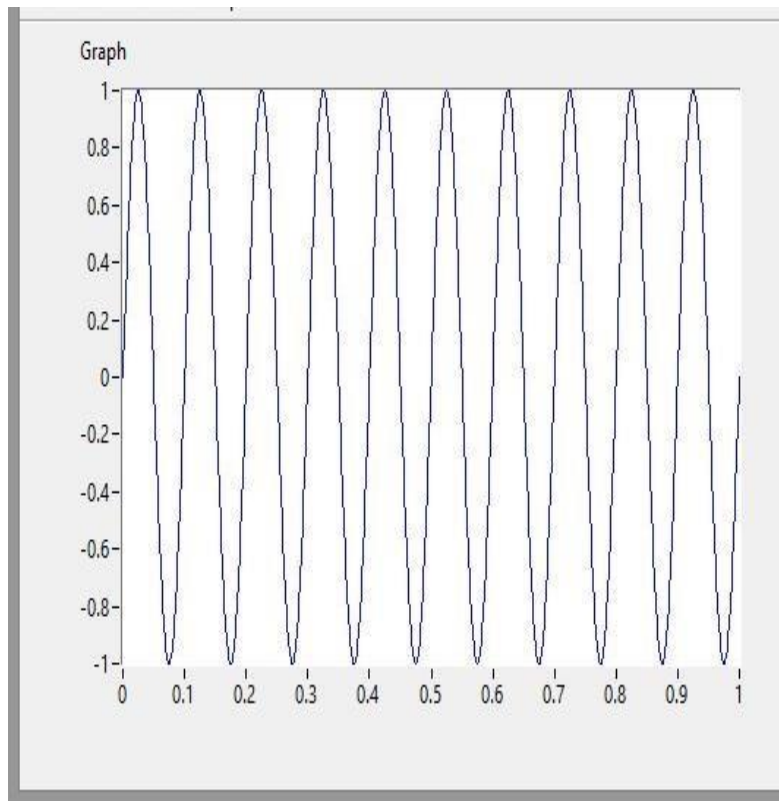
Task 02:

Plot sine wave having 10Hz frequency both in LabVIEW Mathsript and Mathsript node.

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1 fs=1000;
2 f=10;
3 t=0:1/fs:1;
4 x=sin(2*pi*f*t);
5 plot(t,x);
6

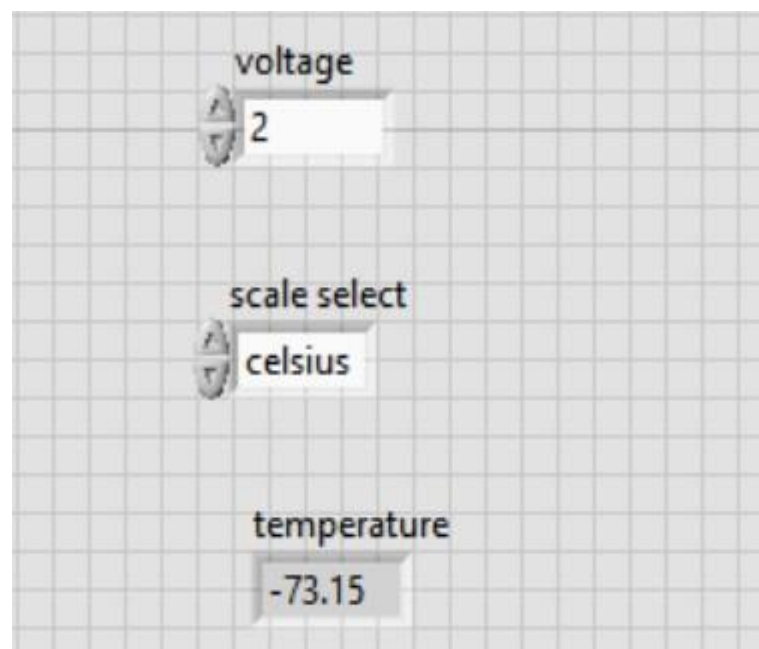
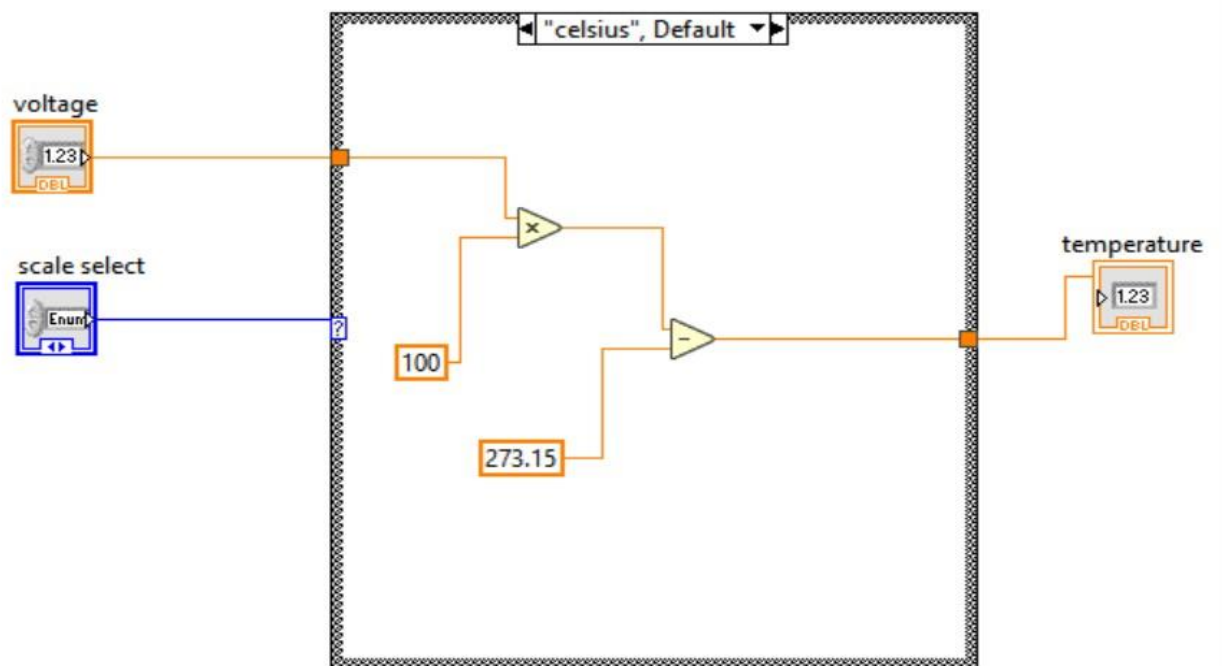
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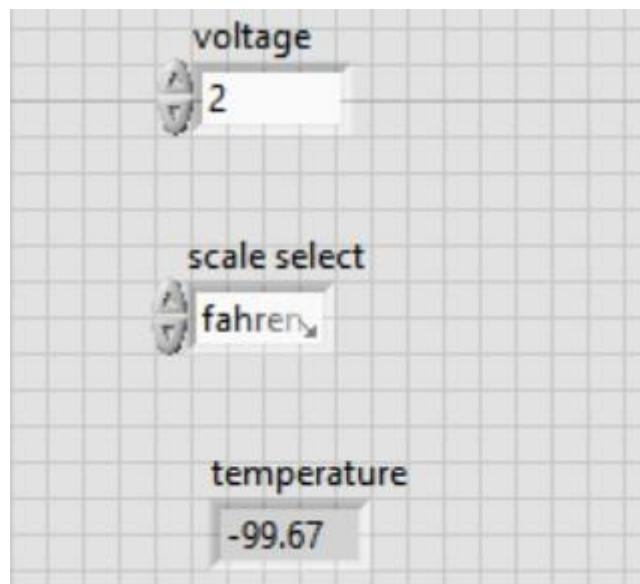
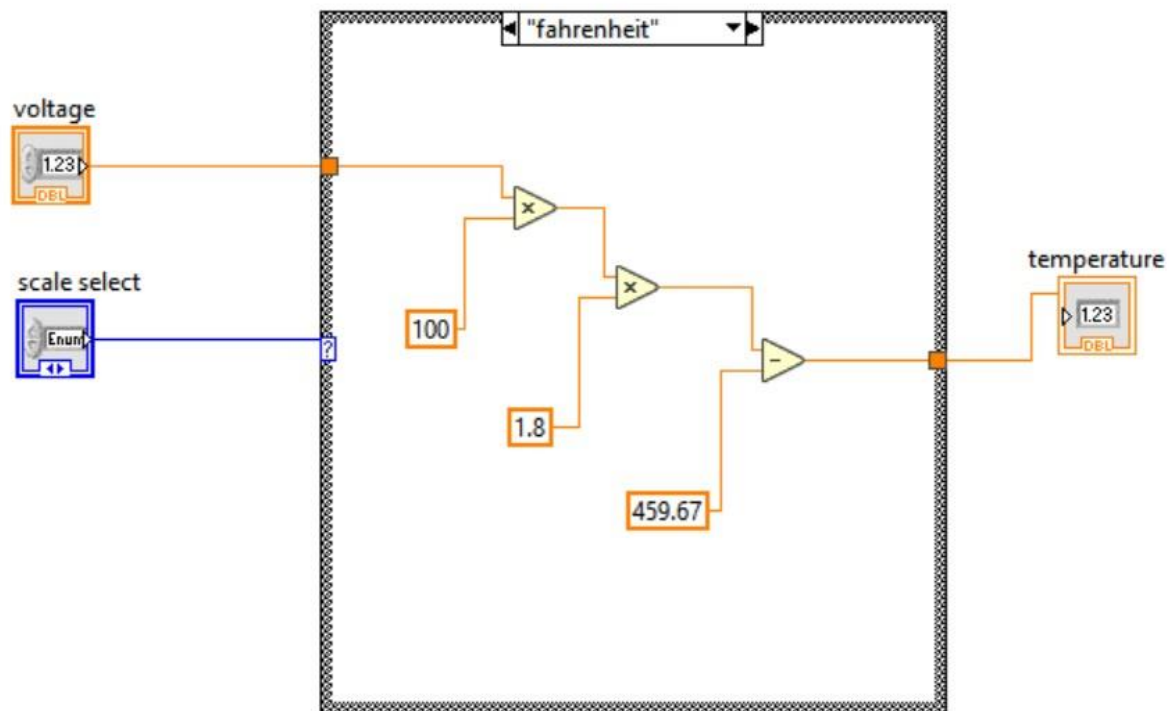
Task 03:

Use the case structure and enum to implement the temperature conversion.

For Celsius:



For Fahrenheit:



Conclusion

This lab introduced the fundamental programming concepts of LabVIEW. Array Control was successfully used to accept user-defined array inputs instead of constant arrays. A 10 Hz sine wave was generated and displayed using both the MathScript window and the MathScript Node, demonstrating different approaches for mathematical computation in LabVIEW. In addition, the Case Structure and Enum were implemented to perform accurate temperature conversion between Celsius and Fahrenheit. Overall, the lab enhanced understanding of LabVIEW programming, graphical data flow, and the use of modern software tools for engineering applications.